

The Basics of AI Chatbots: **Understanding GPT Models and Prompt Engineering**

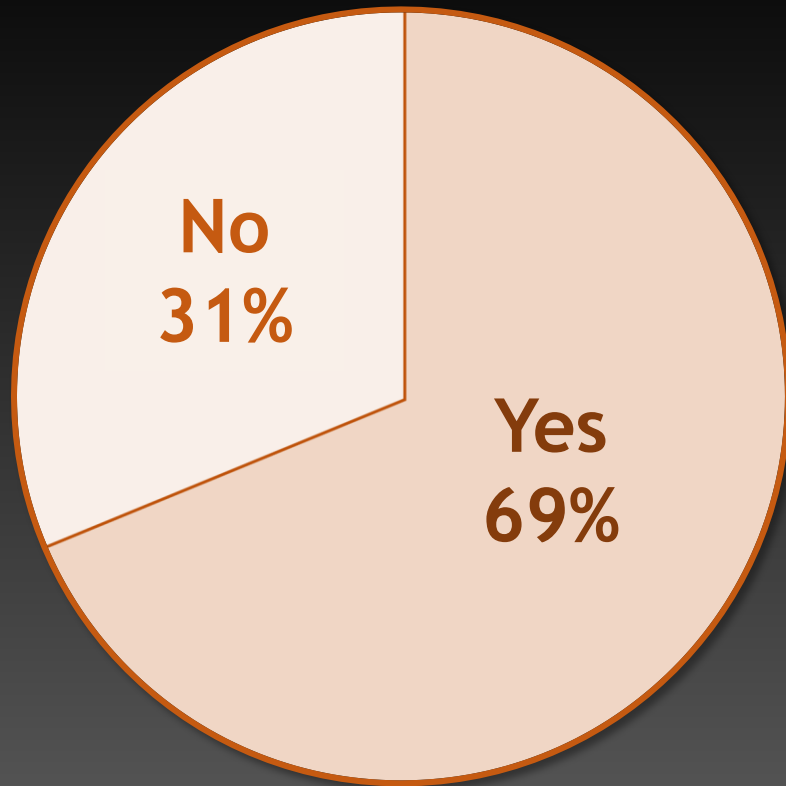


2024 AI Seminar Presenter:

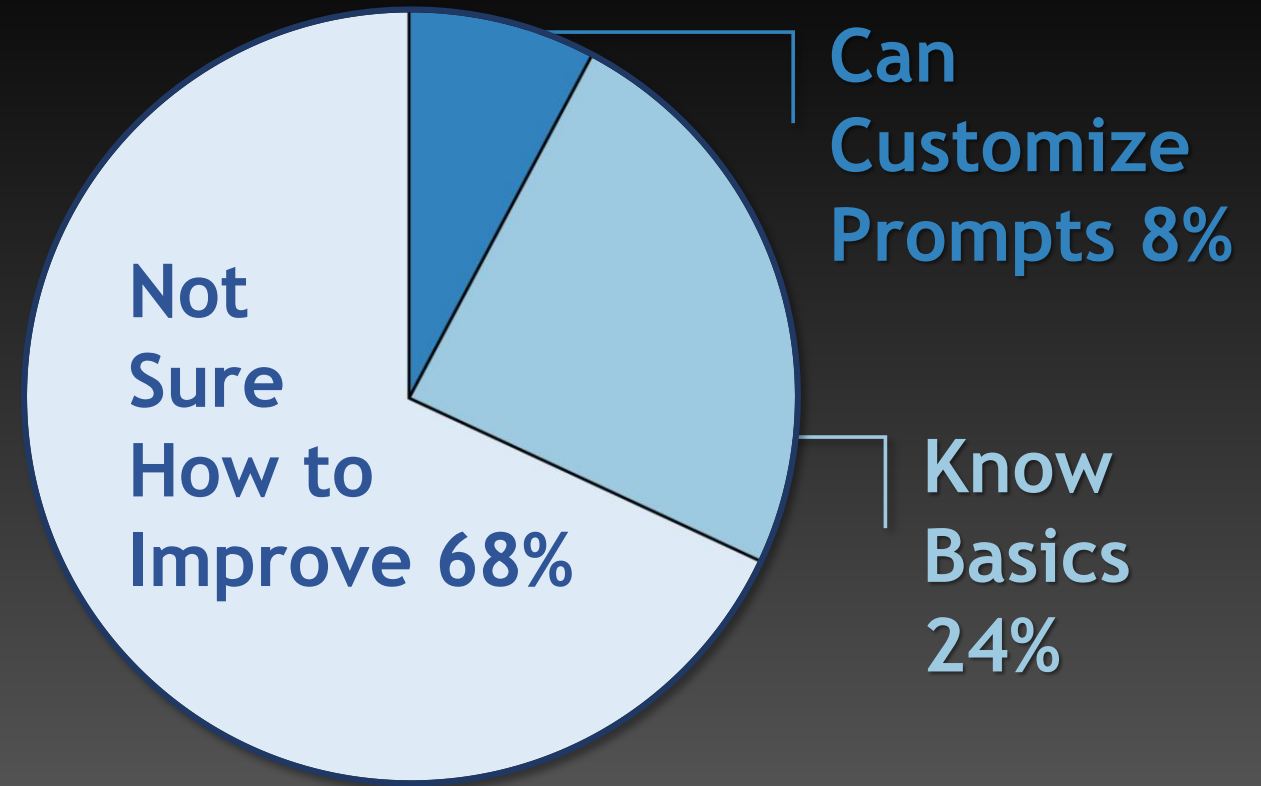
Yuanyuan Wang, PhD

Pre-seminar Survey

AI Chatbot Usage



Prompt Experience



Thank you for your participation!

The Ecosystem of AI: Where does ChatGPT fit?

Root

Branch

Branch

Leaf

ChatGPT:

A specific implementation of an LLM,
optimized for **conversational tasks**.

Large Language Models (LLMs):

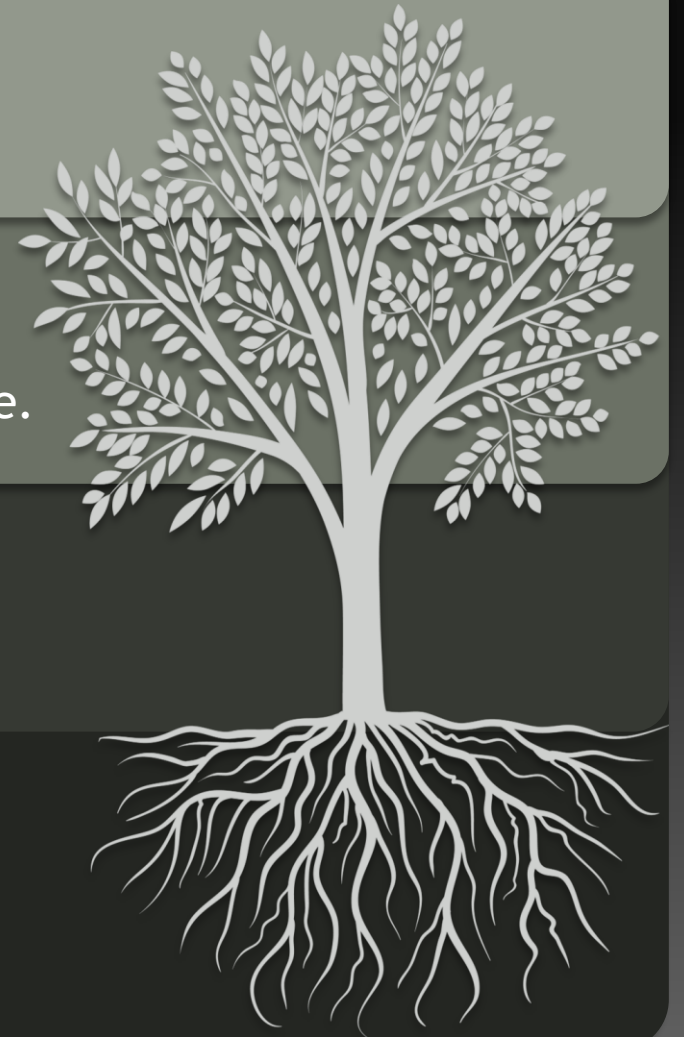
A subset of GenAI,
specialized in processing and generating human language.

Generative AI (GenAI):

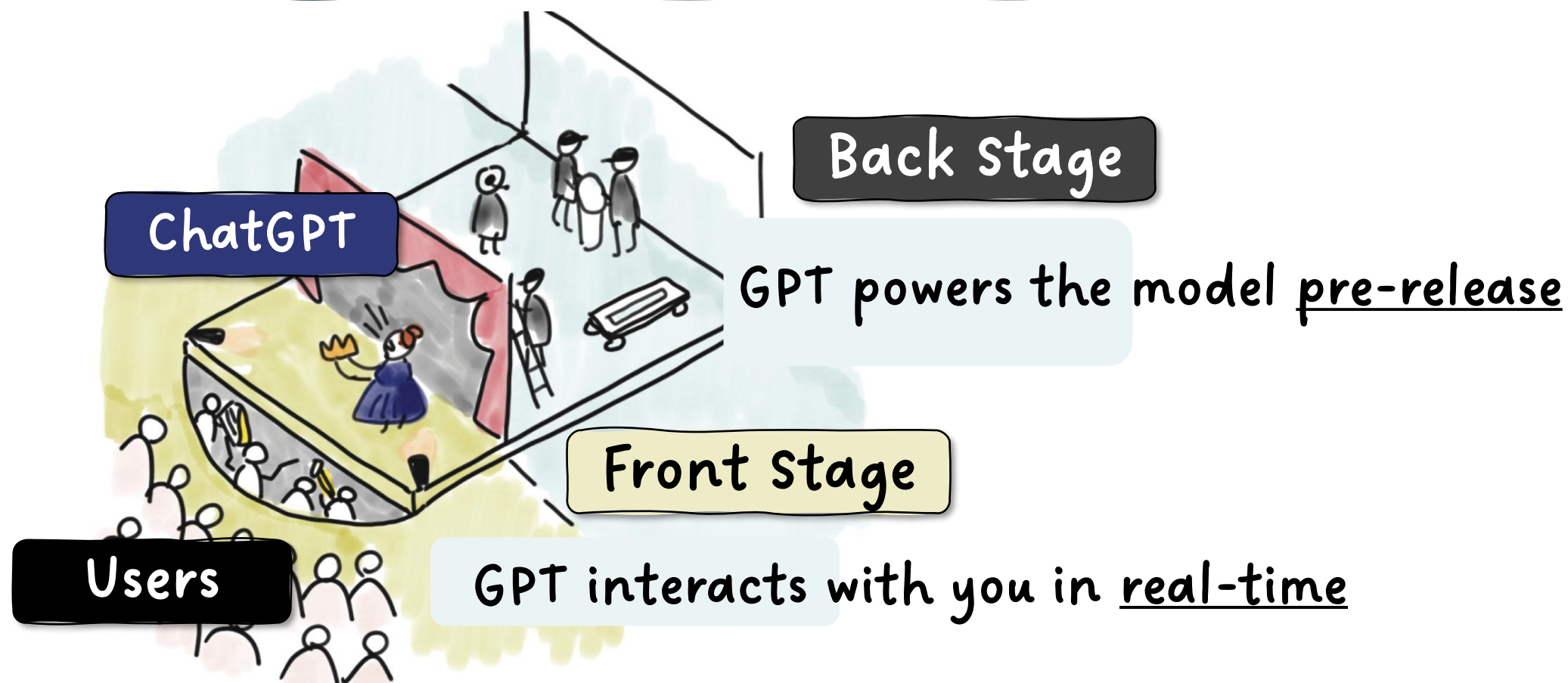
A branch of AI focused on creating new content.

Machine Learning:

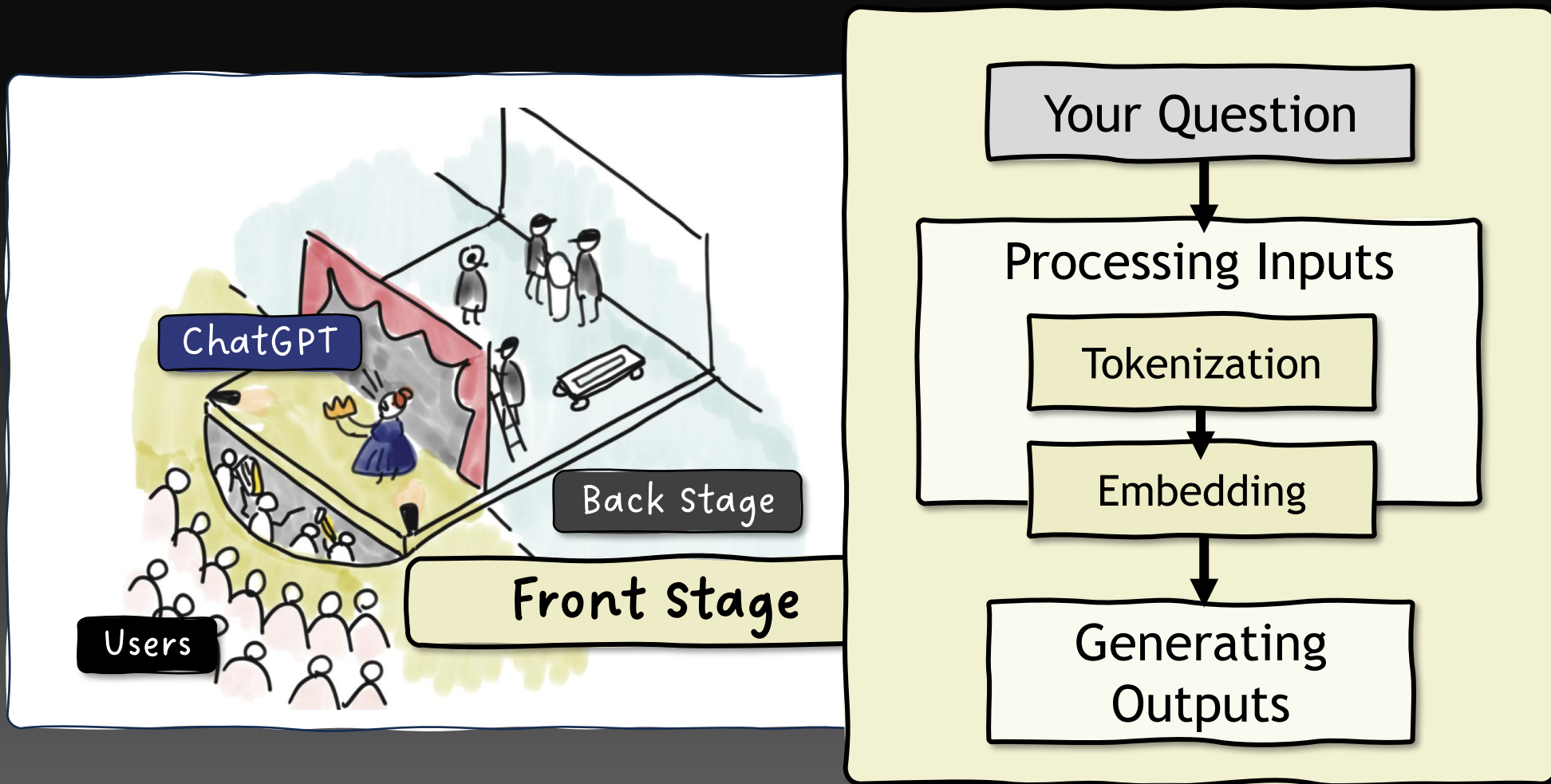
The foundation of modern AI, enabling systems to learn from data.



You'll learn how ChatGPT operates:
on the stage vs. behind the scenes



How ChatGPT Interacts with You In Real Time



ChatGPT Process User Inputs in Tokens

1

“Could you help me create a presentation?”

Could you help me create a presentation? ?

Tokenization

206070

481

1652

668

2501

261

19580

30

Token IDs

Try it: <https://platform.openai.com/tokenizer>

ChatGPT Process User Inputs in **Tokens**



How many e are in the word gene?



There is only one "e" in the word "gene."

Because LLMs see tokens.
Not words.

Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

Jason Wei Xuezhi Wang Dale Schuurmans Maarten Bosma
Brian Ichter Fei Xia Ed H. Chi Quoc V. Le Denny Zhou

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2022



Technique #1 Decomposition Prompting



Let's think step by step.



The word "gene" has four letters: G, E, N, and E.

To determine how many "e"s are in the word, we can go through each letter and count how many of them are "e."

Starting with the first letter, G, there is no "e" in it.

Moving on to the second letter, E, there is one "e" in it.

Next is the third letter, N, which does not contain an "e."

Finally, we have the fourth letter, E again. This also contains one "e."

Adding up the number of "e"s from each letter gives us a total of two "e"s in the word "gene."

So, to answer your question: There are two "e"s in the word "gene."

Chain of Thought (CoT)
Breaks down logical reasoning or
multi-step tasks.

Embeddings Place Tokens in a Coordinate System

1

“Could you help me create a presentation?”

Could you help me create a presentation? ?

Tokenization

206070

481

1652

668

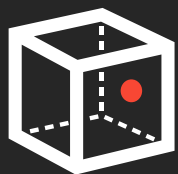
2501

261

19580

30

2



Embedding

0.0390
-0.0123
-0.0208
...

-0.0558
0.0151
0.0031
...

-0.0127
0.0079
0.0781
...

0.7890
-0.0523
-0.0221
...

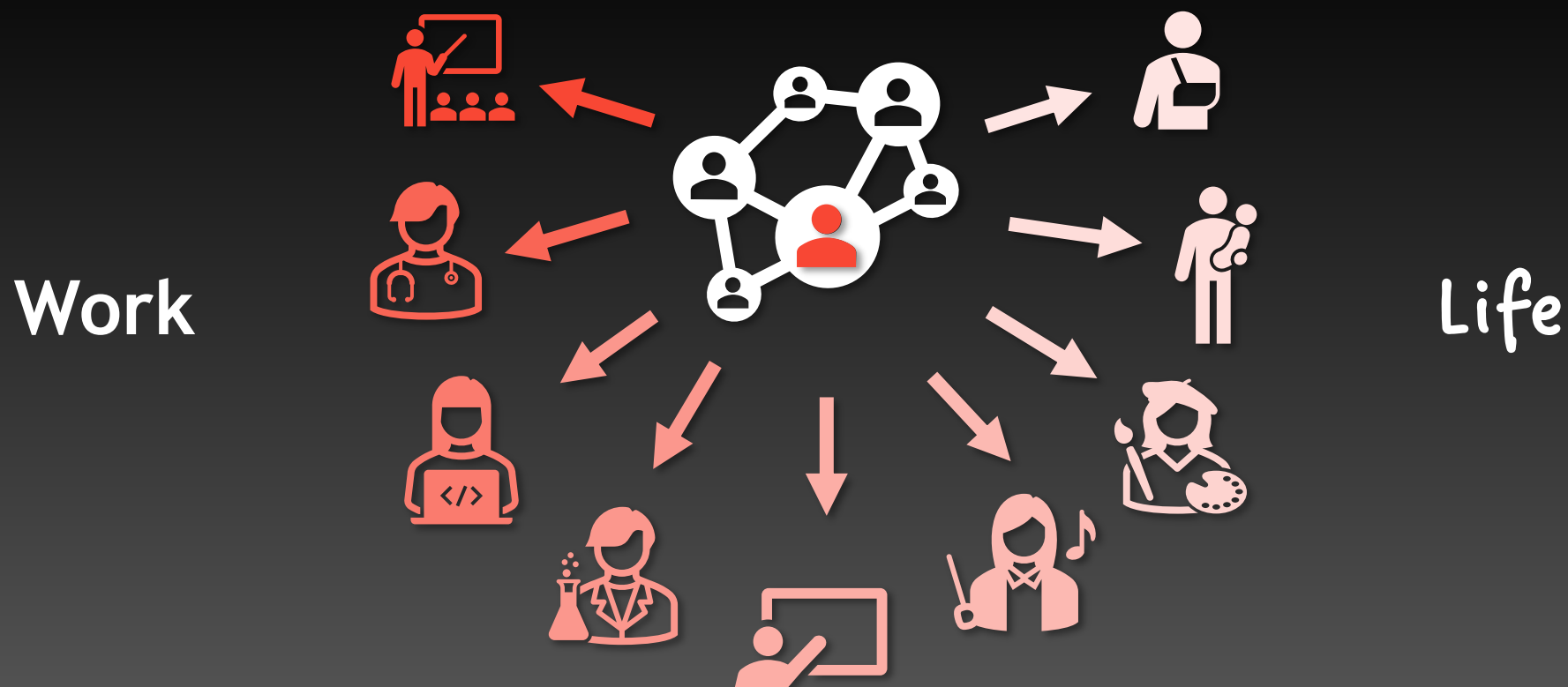
-0.0558
0.0083
0.0931
...

0.1215
-0.2374
-0.8981
...

0.0578
-0.0334
-0.0113
...

-1.1574
0.3824
0.0046
...

Embeddings Aren't Static

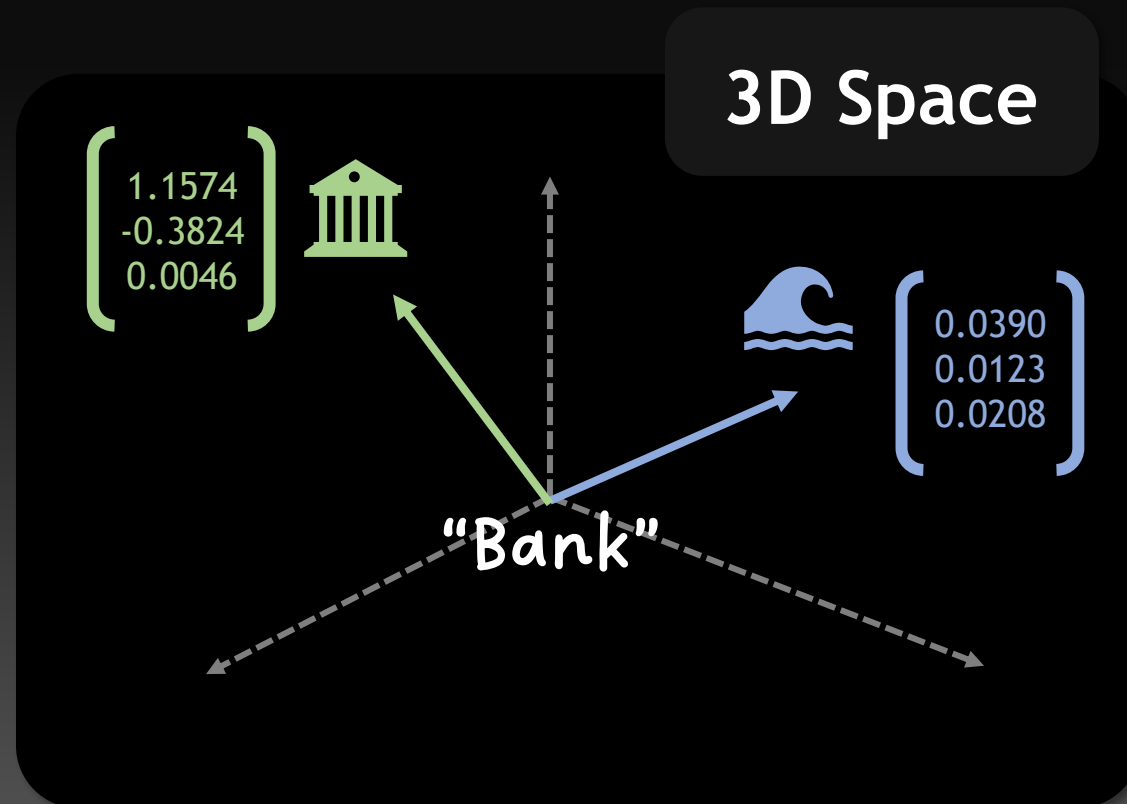


Embeddings Adapt Based on Surrounding Tokens

Contextual Understanding

🌊 I sat on the bank of the river.

🏛️ I deposited money at the bank.



GPT-3 had embeddings with 12,288 dimensions!

**The language of prompting:
What linguistic properties make a prompt successful?**

Alina Leidinger*, Robert van Rooij and Ekaterina Shutova
Institute for Logic, Language and Computation, University of Amsterdam

2023



Technique #2 Specificity Prompting

Persona + Context + Tasks



Sets a clear role

- You are a **reviewer** for a research proposal.
- The **proposal is about** designing a diagnostic test for A using B technology.
- Please **provide a critique** of the proposal, **focusing on** its scientific merit, feasibility, and potential impact.
- **Highlight** any strengths, weaknesses, and areas that need clarification or improvement.
- Here is the proposal [Insert Proposal...]

Gives context

Specifies metrics

Sets output structure

ChatGPT Generate Response Token by Token

User Question:

Could you help me create a presentation?

3

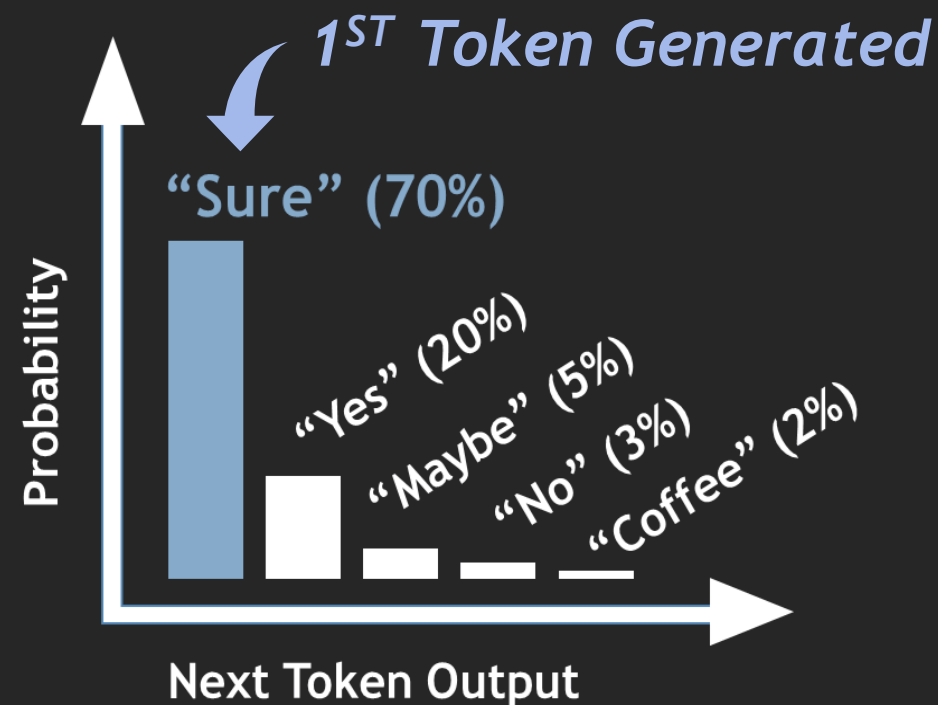
Generate Like
Auto-complete

ChatGPT Generate Response Token by Token

User Question:
Could you help me create a presentation?

3

Generate Like
Auto-complete



ChatGPT Generate Response Token by Token

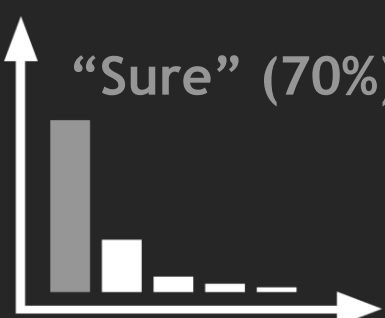
User Question:
Could you help me create a presentation?

3

Generate Like
Auto-complete

1ST Token

"Sure" (70%)



2nd Token Generated

"I" (65%)

"we" (25%)

"can" (9%)

"orange" (1%)

Probability

Next Token Output

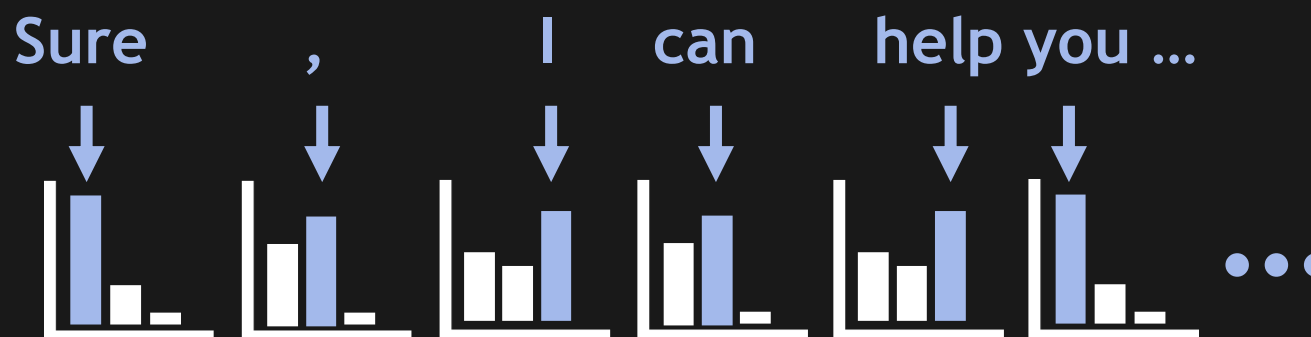
ChatGPT Generate Response Token by Token

User Question:

Could you help me create a presentation?

3

Generate Like
Auto-complete



LEAST-TO-MOST PROMPTING ENABLES COMPLEX REASONING IN LARGE LANGUAGE MODELS

Denny Zhou^{†*} Nathanael Schärli[†] Le Hou[†] Jason Wei[†] Nathan Scales[†] Xuezhi Wang[†]
Dale Schuurmans[†] Claire Cui[†] Olivier Bousquet[†] Quoc Le[†] Ed Chi[†]
[†]Google Research, Brain Team

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Technique #3 Least to Most Prompting



Could you **start with an overview** of the steps to create an R package, then **progressively add more detail**?



- High-level Overview: ...
- Detailed Steps: ...
- In-Depth Explanation: ...

- **Clarity:** avoid overwhelming details
- **Efficiency:** refine based on needs
- **Applicability:** Idea for complex questions

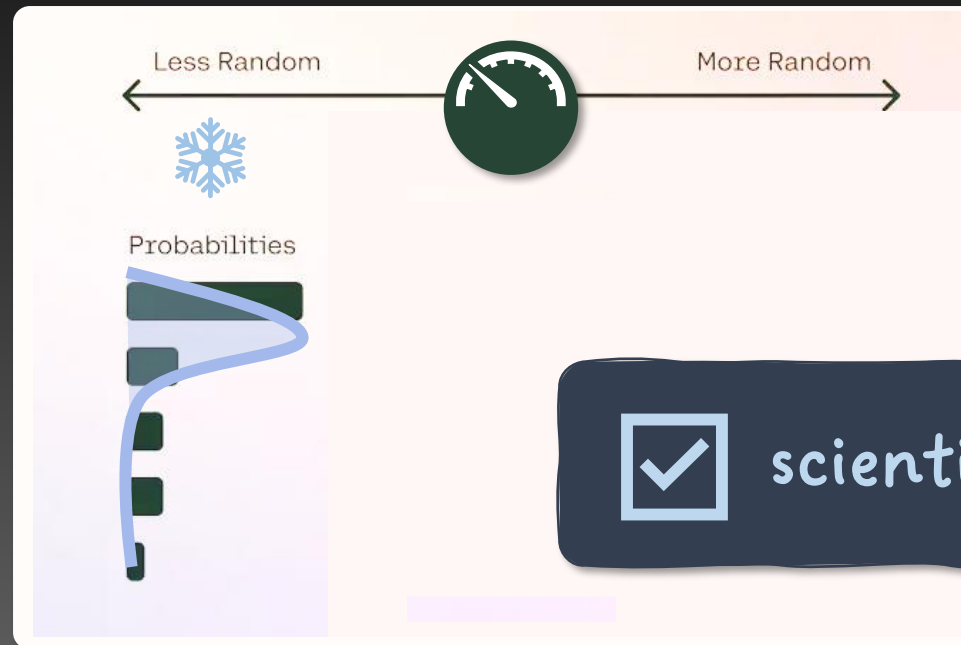
Tuning Creativity in ChatGPT: Temperature

- A parameter that controls output **randomness** and **creativity**.
 - Range: typically, between 0.0 and 1.0.



Tuning Creativity in ChatGPT: Temperature

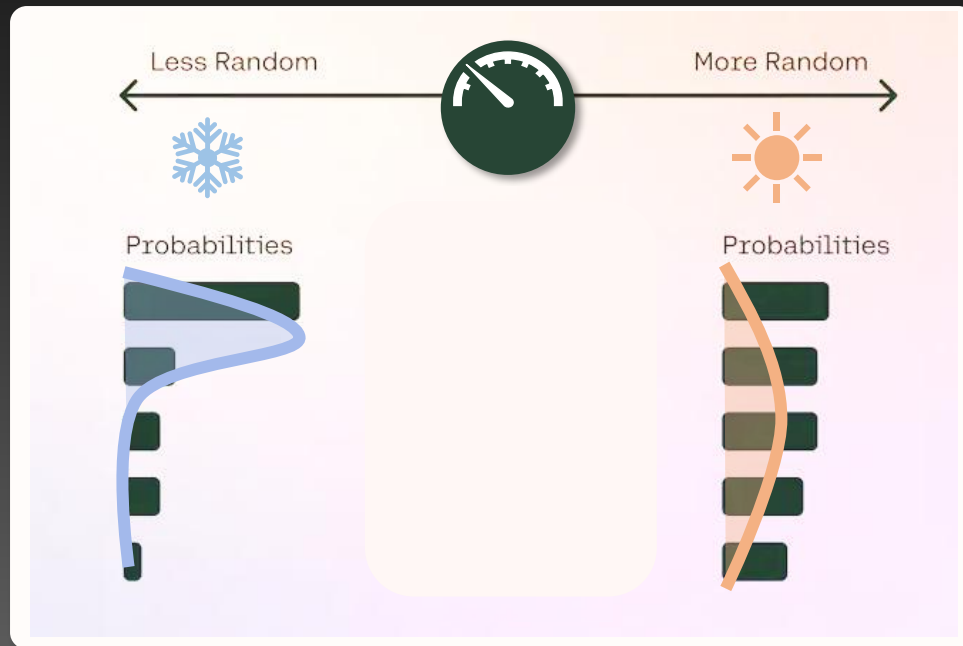
- A parameter that controls output randomness and creativity.
 - Range: typically, between 0.0 and 1.0.



- Low Temperature (Default): Predictable and focused responses.

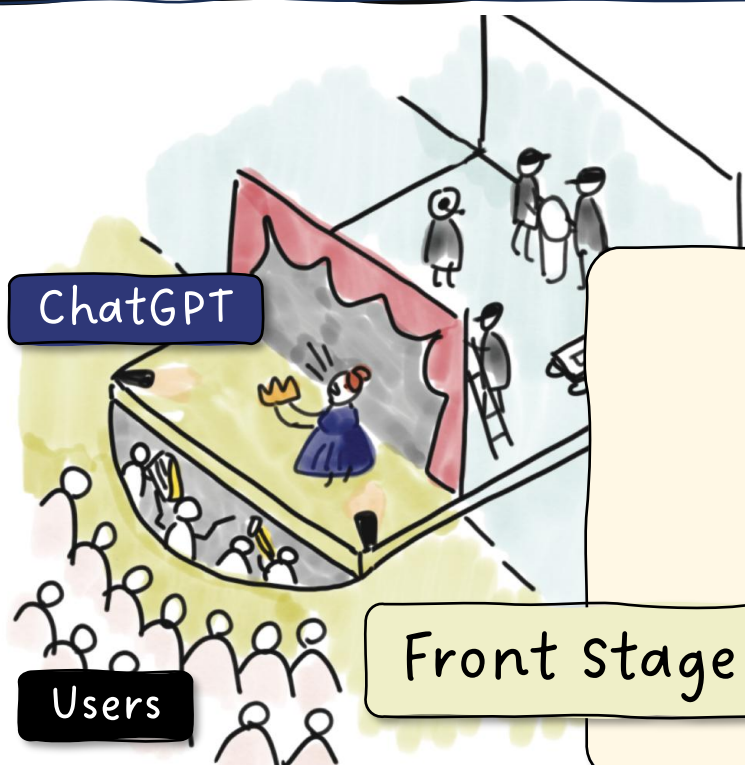
Tuning Creativity in ChatGPT: Temperature

- A parameter that controls output randomness and creativity.
- Range: typically, between 0.0 and 1.0.



- Low Temperature (Default): Predictable and focused responses.
- High Temperature (≥ 0.8): Varied and creative responses.

Guiding ChatGPT: Techniques Summary



1. Chain of Thought: break down tasks into steps



2. Persona + Context + Tasks



3. Least to Most: structure tasks into layers

Behind the Scenes: How ChatGPT Powers Its Performance

ChatGPT

Overview

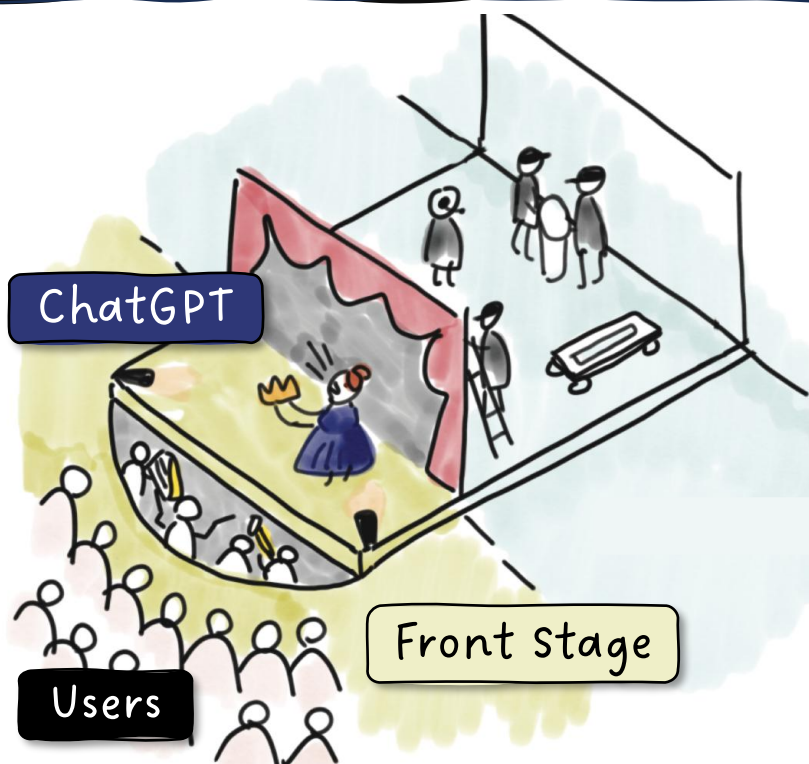
Tokenization

Embeddings

Generation

Temperature

Summary



Back stage

- Transformer Architecture
- Fine-tuning with Human Feedback
- Role of Training Data

Transformer Architecture: The Brain of GPT



- Process text in parallel
- Handles long-range dependencies
- Knows where to focus

Attention Is All You Need

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2017

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Fine-Tuning with Human Feedback



Training language models to follow instructions with human feedback

Long Ouyang* Jeff Wu* Xu Jiang* Diogo Almeida* Carroll L. Wainwright*
Pamela Mishkin* Chong Zhang Sandhini Agarwal Katarina Slama Alex Ray
John Schulman Jacob Hilton Fraser Kelton Luke Miller Maddie Simens
Amanda Askell† Peter Welinder Paul Christiano*†
Jan Leike* Ryan Lowe*

OpenAI

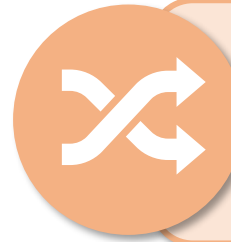
Reinforcement Learning with Human Feedback (RLHF)

- Human evaluations on **helpfulness, truthfulness, and harmlessness**
- A reward model was trained to replace human evaluators.

SELF-CONSISTENCY IMPROVES CHAIN OF THOUGHT REASONING IN LANGUAGE MODELS

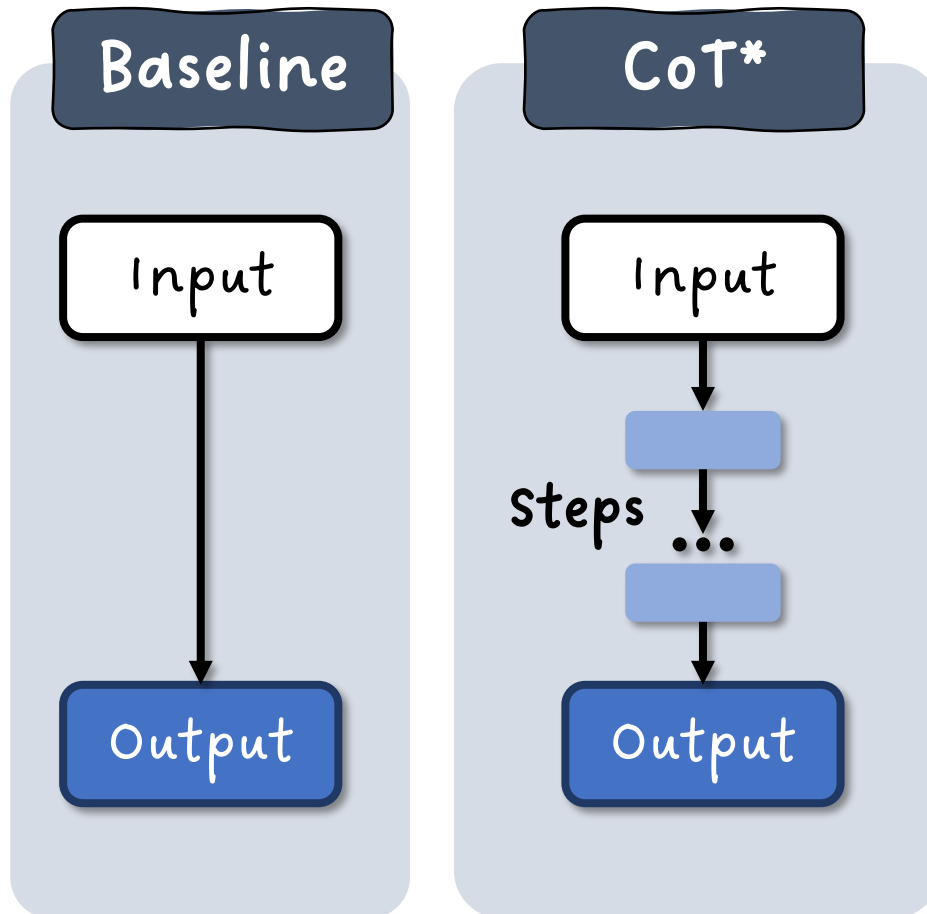
Xuezhi Wang^{†‡} Jason Wei[†] Dale Schuurmans[†] Quoc Le[†] Ed H. Chi[†]
Sharan Narang[†] Aakanksha Chowdhery[†] Denny Zhou^{†§}
[†]Google Research, Brain Team
[‡]xuezhiw@google.com, [§]dennyzhou@google.com

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Technique #4 Self-consistency Prompting

*: Chain of Thought



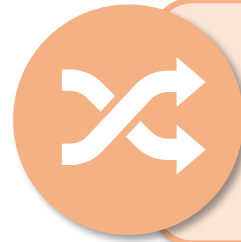
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Xuezhi Wang^{†‡} Jason Wei[†] Dale Schuurmans[†] Quoc Le[†] Ed H. Chi[†]
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[†]Google Research, Brain Team

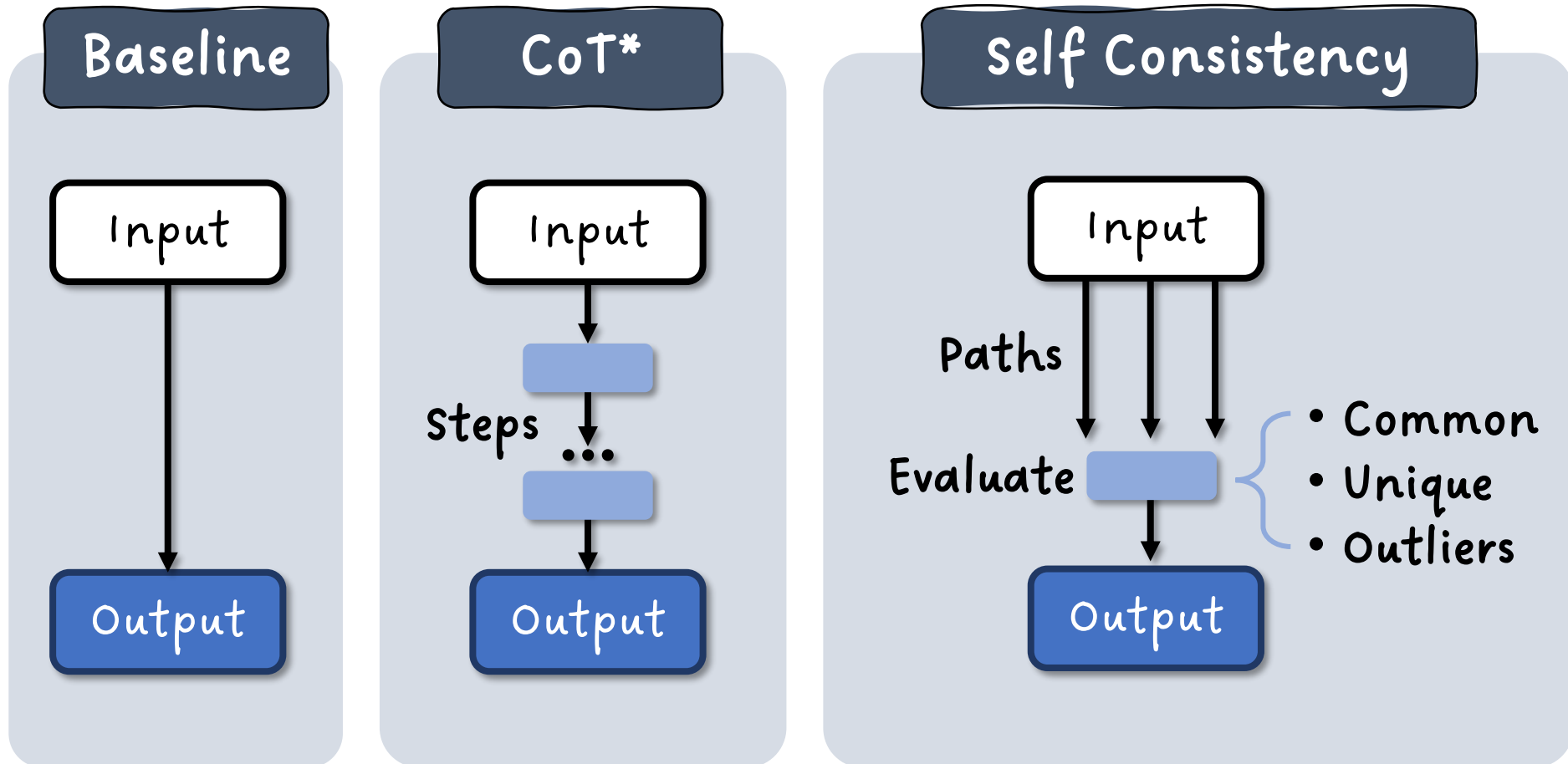
[‡]xuezhiw@google.com, [§]dennyzhou@google.com

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Technique #4 Self-consistency Prompting

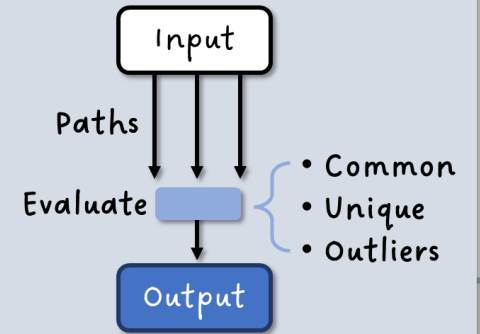
*: Chain of Thought





- How can genomic data help track the spread of an emerging pathogen?
- Generate multiple reasoning paths and synthesize a consistent strategy.

Self Consistency



Generated Paths:

- **Path 1:** Use **phylogenetic analysis** to identify clusters
- **Path 2:** Monitor **mutation frequency** across time and regions
- **Path 3:** Combine genomic data with **epi metadata** to track transmission routes.

Evaluation:

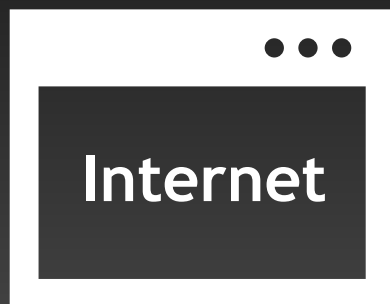
- **Common Elements:** Phylogenetic analysis and mutation monitoring are consistently recommended.
- **Unique Insight:** Metadata integration adds context and precision, though not mentioned in every path.
- **Outliers:** Any overly complex or impractical suggestions will be eliminated.

Refined Response:

To track the spread of an emerging pathogen, use phylogenetic analysis to identify related strains and monitor mutation frequency across time and regions. Integrate genomic data with metadata such as patient demographics and travel history to map transmission routes.

Diverse Training Data: The Foundation of GPT

- Sources of Training Data:



- Web Scraping



WIKIPEDIA

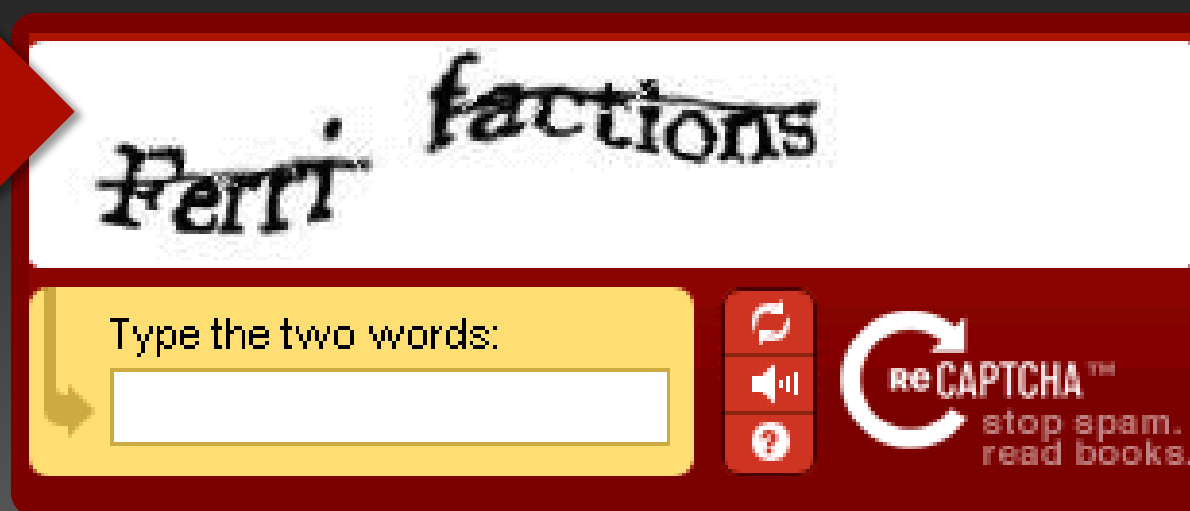
- Wikipedia



- Code Repositories

Diverse Training Data: The Foundation of GPT

- Sources of Training Data:
 - Web Scraping
 - Wikipedia
 - Code Repositories
 - Books and Academic Texts.



The Scale of LLM Training Data

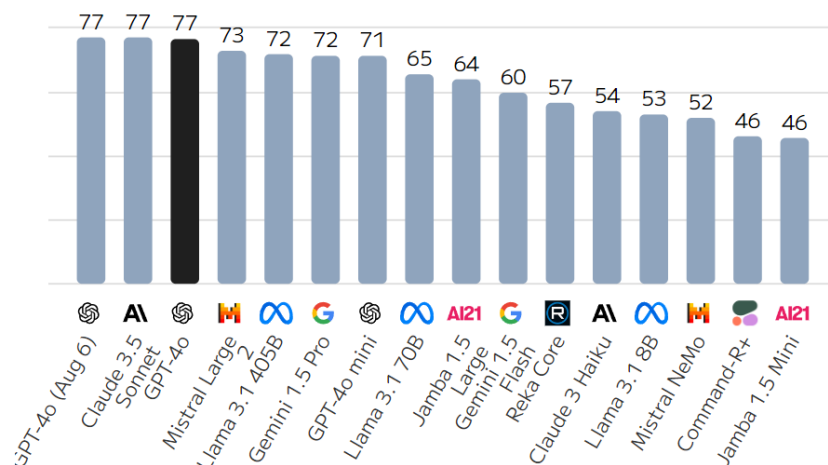
	Languages	Documents	Estimated Tokens
~400 Million Books	English	14.5B	20.5T
	German	1.9B	3.0T
1 Trillion token ~ 15 million books (book ~ 50,000 words)	French	1.6B	2.7T
	Spanish	1.8B	2.8T
	Italian	0.9B	1.5T
Total		20.8B	30.4T

Source: RedPajam-Data-v2:

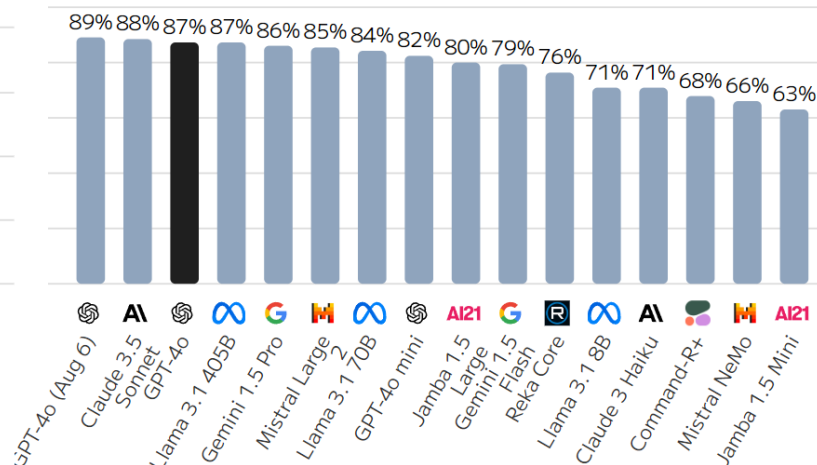
An Open Dataset with 30 Trillion Tokens for LLM training.

Benchmarks on knowledge, reasoning, math, and coding.

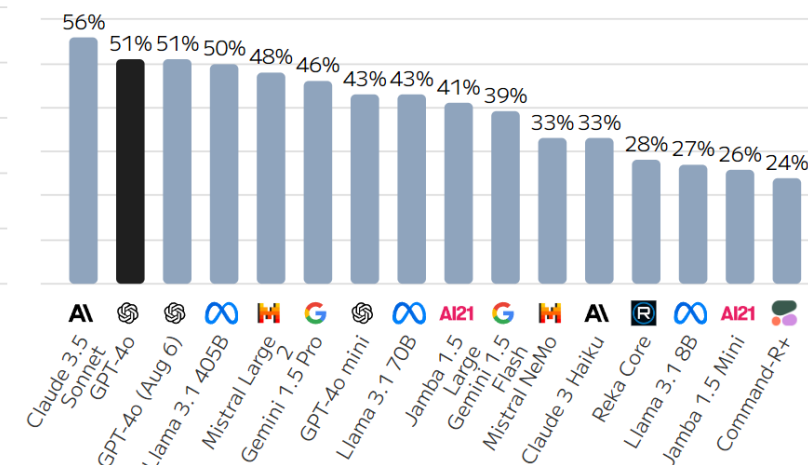
Artificial Analysis Quality Index



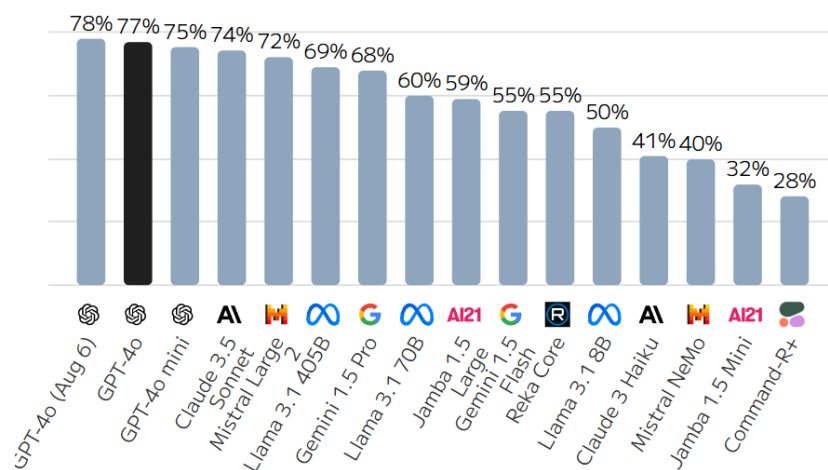
Reasoning & Knowledge (MMLU)



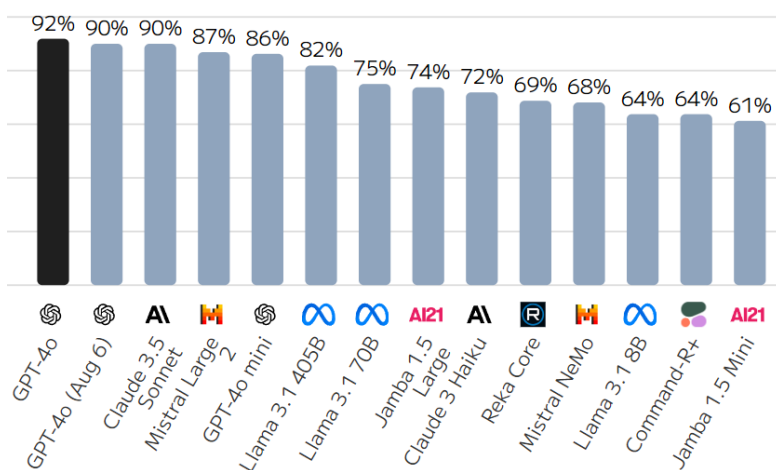
Scientific Reasoning & Knowledge (GPQA)



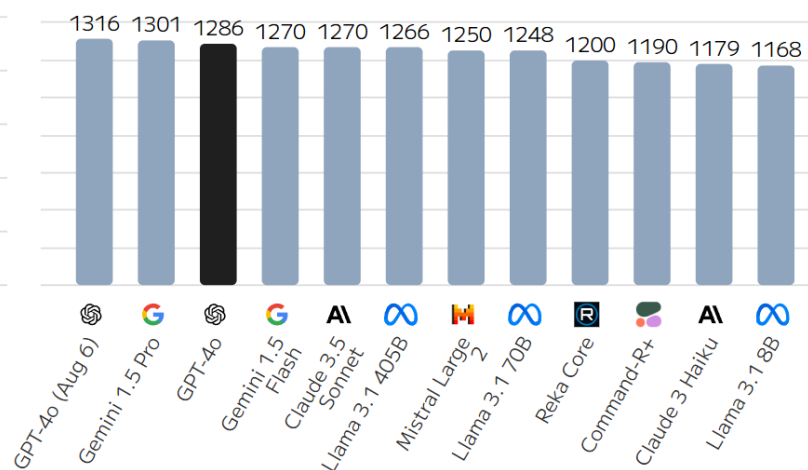
Quantitative Reasoning (MATH)



Coding (HumanEval)



Communication (LMSys Chatbot Arena ELO Score)

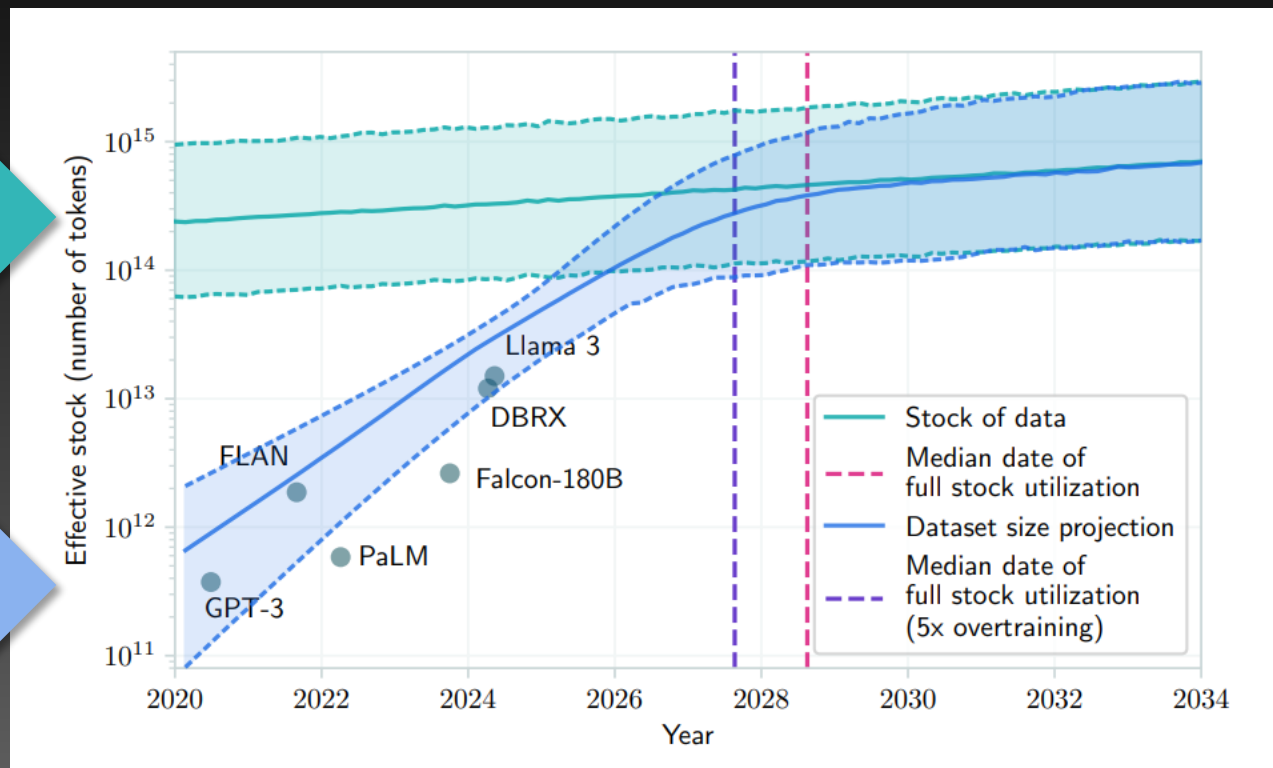


The Data Dilemma: Running Out of Training Data?

- Running Out of Training Data in 2028?
 - Finite Data Pool, Data Redundancy, Quality Over Quantity

Available

Used



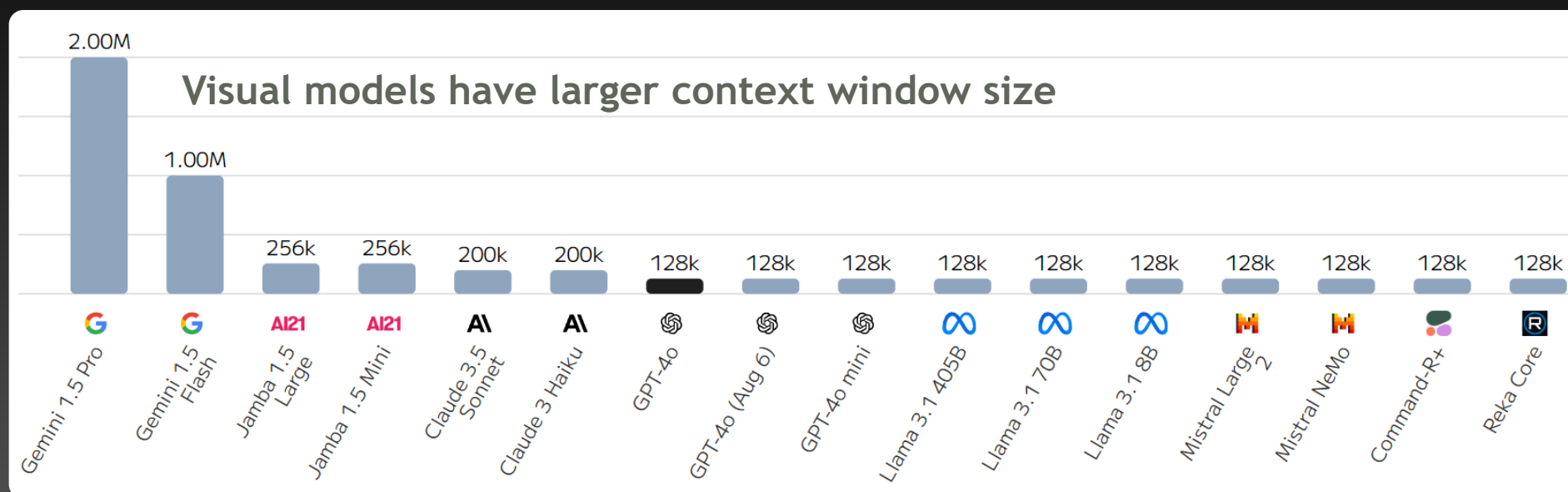
Villalobos, Pablo, et al.

"Will we run out of data? Limits of LLM scaling based on human-generated data."

The 41th International Conference on Machine Learning, 2024.

Context Window Size: What GPT Can “Remember”

- The amount of text ChatGPT can “remember” and consider.
- Theoretical Limit: 128k tokens, or ~96,000 words (a book of 300 pages).
- Reality Check: The real limit is much less than 128k.



[GPT-4o - Quality, Performance & Price Analysis](#) | Artificial Analysis



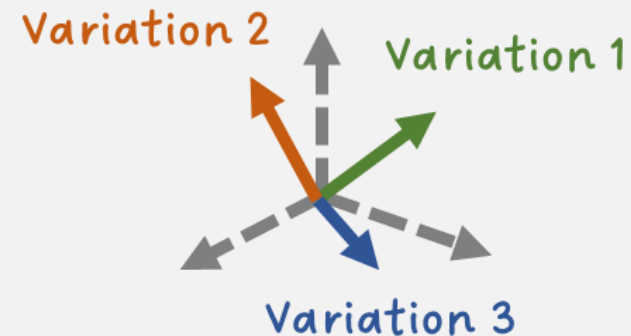
Technique #5 Exploration Prompting



- You are a bioinformatics expert tasked with analyzing a dataset of viral genomes to identify patterns of drug resistance.
- The dataset includes metadata on geographic origin, host species, and collection dates.
- **Generate 3 different analytical strategies and summarize potential strengths.**



Strategy 1: Phylogenetic Analysis: ...
Strategy 2: Regression or machine learning ...
Strategy 3: Network/Clustering Analysis:...



Can Large Language Models Explore In-Context?

Akshay Krishnamurthy^{*1}, Keegan Harris^{†2}, Dylan J. Foster¹,
Cyril Zhang¹, and Aleksandrs Slivkins^{*1}

2024/10

¹Microsoft Research
²Carnegie Mellon University



Technique #5 Exploration Prompting

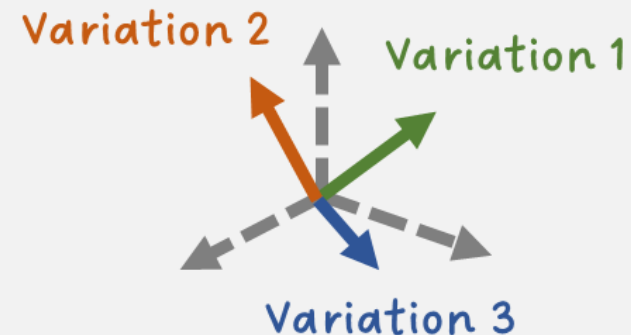
Without substantial interventions,
LLMs do not robustly engage in exploration.



- You are a bioinformatics expert tasked with analyzing a dataset of viral genomes to identify patterns of drug resistance.
- The dataset includes metadata on geographic origin, host species, and collection dates.
- **Generate 3 different analytical strategies and summarize potential strengths.**



Strategy 1: Phylogenetic Analysis: ...
Strategy 2: Regression or machine learning ...
Strategy 3: Network/Clustering Analysis:...



Literature Review Tools

1

Semantic Scholar



SEMANTIC SCHOLAR

Type of site

Search engine

Created by

Allen Institute for Artificial Intelligence

URL

semanticscholar.org

Launched

November 2, 2015; 8 years ago^[1]

Google Scholar

DOI: 10.1007/978-981-97-4581-4_17 • Corpus ID: 271956766

Epidemic Information Extraction for Event-Based Surveillance using Large Language Models

[Sergio Consoli](#), [Peter Markov](#), +3 authors [Mario Ceresa](#) • Published 26 August 2024 •

Computer Science, Medicine, Environmental Science

TLDR A novel approach to epidemic surveillance is presented, leveraging the power of Artificial Intelligence and Large Language Models for effective interpretation of unstructured big data sources, like the popular ProMED and WHO Disease Outbreak News. [Expand](#)



[PDF] Semantic Reader



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Cite

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Ask This Paper

AI-Powered

What is the goal of this paper?

What are the key results of this paper?

What methods are used?

Ask a question that can be answered by this paper.

Literature Review Tools

1

Semantic Scholar



SEMANTIC SCHOLAR

Type of site

Search engine

Created by

Allen Institute for Artificial Intelligence

URL

semanticscholar.org

Launched

November 2, 2015; 8 years ago^[1]

Google Scholar

2



Consensus

Consensus meter

20 papers analyzed

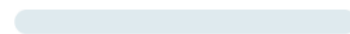
✓ Yes - 90%



— Possibly - 10%



✗ No - 0%



3

PubTator³

NIH NLM

FTP API Tutorial

Home

Saved

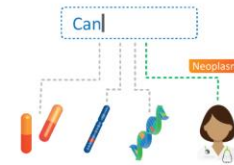
Playlists

Search entities & relations in 35+ million biomedical publications.

Ex: Remdesivir



Try: N-dimethylnitrosamine and Metformin COVID-19 and PON1



Entities Autocomplete

PubTator3 uses a high-performance entities search engine, to normalize different forms of the same entity into a unique standardized name to returned all matching articles.



Full Text Search

PubTator3 provides unified access to the entire 35+ million abstracts in PubMed and nearly 6 million full-text articles in the PMC Text Mining subset.



Relations

PubTator3 allows to filter results to only return publications containing specific relations between two entities, such as diseases, chemicals, genes or variants.

Bio-concepts and
their frequencies:



Home	
Saved	
Playlists	
BIOCONCEPTS AND MENTIONS	
GENE	
SPIKE	16
RDRP	6
ANGIOTENSIN-CONVERTING ENZYME 2	3
NUCLEOCAPSID	2
E	1
more	
DISEASE	
COVID-19	20
INFECTION	10
BA.2	10
VUM	9
BA.5	4
more	
CHEMICAL	
BA.1	2
BAMLANIVIMAB	1
CASIRIVIMAB	1
VARIANT	
E484K	12
L452R	7
D614G	6
K417N	6
N501Y	4
N439K	4
P681R	2
T478K	2
69-70DEL	2
F486V	2
S477G	2
F486S	2
P681H	1
K417T	1
E484A	1

PMID37539393 • PMC10394493

Jul 3, 2023

Emergence of COVID-19 Variants: An Update

Gupta P, Gupta V ... Singhal L • Cureus

[Save](#) [Share](#) [Cite](#) [Download](#) [Add To Playlist](#)

Severe acute respiratory disease virus-2 (SARS CoV-2) is one of the deadliest global threats faced by mankind to date. Despite the colossal efforts, the viral pandemic swept across all boundaries. Besides the virulence and susceptible population, the low proofreading capacity and error-prone mechanism of RNA-dependent RNA polymerase (RdRp) have contributed to new variants and reinfections. The World Health Organization has officially categorized these variants as variants of concern or variants of interest. This nomenclature is not merely to suffice the surveillance but also to have effective treatment and vaccine options in place. Coronavirus disease 2019 (COVID-19) variants have the propensity to render the available treatment strategies futile owing to the mutations they acquire. The futility of treatment strategies can be attributed either to the ineffectiveness or the shortage of supply given the skyrocketing increase in the number of cases. Presently, the Omicron variant is the most widespread one and is known to escape the protection, be it immune-derived, vaccination-derived, or hybrid. WHO has recommended modification in vaccine development policies and few companies have introduced Omicron-adapted vaccine jabs. Keeping in view the unending tale of COVID-19 variants and the huge data available on the same, this review focuses on providing insight into the emergence and ongoing dynamics of these new COVID-19 variants.

Introduction and background

Severe acute respiratory disease virus-2 (SARS-CoV-2) is one of the most dreadful threats to mankind and has culminated into an unrelenting, fatal pandemic. The virus emerged in Wuhan, China, in late 2019 affecting a cluster of patients with pneumonia of unknown etiology. Consequently, the infection was affirmed as a 'public health emergency of international concern' on January 30, 2020, by the World Health Organization (WHO). The origin of the virus was traced back to the wet live markets of China, which facilitated the zoonotic transmission of disease by spillover across the animal-human interface. The virus gradually gripped the whole world, and WHO acknowledged it as a global pandemic on March 11, 2020. Genome sequencing of the agent unraveled the virus to be a member of the order

SHOW BIOCONCEPTS

☒ GENE ☒ DISEASE
☒ CHEMICAL ☒ VARIANT
☒ SPECIES ☒ CELLLINE

OPTIONS

☐ SHOW ABSTRACT

Sections

TITLE

INTRODUCTION AND BACKGROUND

REVIEW

CONCLUSIONS

REFERENCES

5 Prompt Techniques



1. Chain of Thought



2. Persona + Context + Task



3. Least to Most



4. Self-Consistency



5. Explore Diverse Response

Key Takeaways

“The real question is not whether machines think but whether men do.”

B.F. Skinner

- **GPT Core mechanisms:**
 - Tokenization
 - Embeddings
 - Transformer
 - Fine-tuning with human feedback
 - Scale of Training Data
- **Challenges Ahead:**
 - Context window size
 - Data availability
- **Looking Forward:**
 - Code generation
 - Data visualization